

解: $\max Z = -5x_1 + 5x_2 + 13x_3$

$$\begin{cases} -x_1 + x_2 + 3x_3 + x_4 = 20 \\ 12x_1 + 4x_2 + 10x_3 + x_5 = 90 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_4 & 0 & 20 & -1 & 1 & 3 & 1 & 0 & \frac{20}{3} \\ x_5 & 0 & 90 & 12 & 4 & 10 & 0 & 1 & 9 \end{array}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_3 & 13 & \frac{20}{3} & -\frac{1}{3} & \frac{1}{3} & 1 & \frac{1}{3} & 0 & 20 \\ x_5 & 0 & \frac{70}{3} & \frac{16}{3} & \frac{2}{3} & 0 & -\frac{10}{3} & 1 & 35 \end{array}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_2 & 5 & 20 & -1 & 1 & 3 & 1 & 0 & \\ x_5 & 0 & 10 & 16 & 0 & -2 & -4 & 1 & \\ & & 0 & 0 & -2 & -5 & 0 & & \end{array}$$

$\therefore$ 最优解为 $(0, 20, 0, 0, 10)^T$, $Z_{\max} = 100$

(1)
$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_2 & 5 & 30 & -1 & 1 & 3 & 1 & 0 & \\ x_5 & 0 & -30 & 16 & 0 & -2 & -4 & 1 & \\ & & 0 & 0 & -2 & -5 & 0 & & \end{array}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_2 & 5 & -15 & 23 & 1 & 0 & -5 & \frac{3}{2} & \\ x_5 & 13 & 15 & -8 & 0 & 1 & 2 & -\frac{1}{2} & \\ & & -16 & 0 & 0 & -1 & -1 & & \end{array}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_4 & 0 & 3 & -\frac{23}{5} & -\frac{1}{5} & 0 & 1 & -\frac{3}{10} & \\ x_3 & 13 & 9 & \frac{6}{5} & \frac{2}{5} & 1 & 0 & \frac{1}{10} & \\ & & -\frac{103}{5} & -\frac{1}{5} & 0 & 0 & -\frac{13}{10} & & \end{array}$$

$\therefore$ 最优解为 $(10, 0, 9, 3, 0)^T$, $Z_{\max} = 117$

(2)
$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_4 & 5 & 20 & -1 & 1 & 3 & 1 & 0 & \\ x_5 & 0 & -10 & 16 & 0 & -2 & -4 & 1 & \\ & & 0 & 0 & -2 & -5 & 0 & & \end{array}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & & \\ x_2 & 5 & 5 & 23 & 1 & 0 & -5 & \frac{3}{2} & \\ x_3 & 13 & 5 & -8 & 0 & 1 & 2 & -\frac{1}{2} & \\ & & -16 & 0 & 0 & -1 & -1 & & \end{array}$$

地 最优解为 $(0, 5, 5, 0, 0)^T$, $Z_{\max} = 90$

(3) 更改 c , b 未改变, σ 未变

最优解不变

(4) $\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$

$$\sigma'_1 = -5 - (5 \ 0) \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 5 \end{pmatrix} = -5$$

$\therefore$ 最优解不变

(5)
$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & 0 & \\ x_4 & 5 & 20 & -1 & 1 & 3 & 1 & 0 & 0 \\ x_5 & 0 & 10 & 16 & 0 & -2 & -4 & 1 & 0 \\ x_6 & 0 & 50 & 2 & 3 & 5 & 0 & 0 & 1 \end{array}$$

$$\begin{array}{ccccccccc} & & -5 & 5 & 13 & 0 & 0 & 0 & \\ x_2 & 5 & 20 & -1 & 1 & 3 & 1 & 0 & 0 \\ x_5 & 0 & 10 & 16 & 0 & -2 & -4 & 1 & 0 \\ x_6 & 0 & -10 & 5 & 0 & -4 & -3 & 0 & 1 \end{array}$$

$$\begin{array}{ccccccccc} & & 0 & 0 & -2 & -5 & 0 & 0 & \\ & & -5 & 5 & 13 & 0 & 0 & 0 & \\ x_2 & 5 & \frac{25}{2} & \frac{1}{2} & 0 & -\frac{5}{2} & 0 & \frac{1}{2} & \\ x_5 & 0 & 15 & \frac{31}{2} & 0 & 0 & -\frac{5}{2} & 1 & -\frac{1}{2} \\ x_3 & 13 & \frac{15}{2} & -\frac{5}{2} & 0 & 1 & \frac{3}{2} & 0 & -\frac{1}{2} \end{array}$$

$\therefore$ 最优解为 $(0, \frac{25}{2}, \frac{5}{2}, 0, 15)^T$, $Z_{\max} = 95$

(6) $P'_1 = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 10 \end{pmatrix} = \begin{pmatrix} -1 \\ 14 \end{pmatrix}$

$$P'_2 = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\sigma'_1 = -5 - (5 \ 0) \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 10 \end{pmatrix} = 0$$

$$\sigma'_2 = 5 - (5 \ 0) \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 5 \end{pmatrix} = 0$$

$$b' = \begin{pmatrix} 20 \\ 10 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 10 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \end{pmatrix}$$

$\therefore$ 最优解不变

地 最优解为 $(0, 5, 5, 0, 0)^T$, $Z_{\max} = 90$

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